

PVSYST V6.43					19/06/26	Page 1/4
Grid-Connected System: Simulation parameters						
Project :		Graduation project				
Geographical Site		Graduation_Egypt		Country	Egypt	
Situation		Latitude	30.0°N	Longitude	31.8°E	
Time defined as		Legal Time	Time zone UT+2	Altitude	310 m	
		Albedo	0.20			
Meteo data:		Graduation_Egypt		Meteonorm 7.1 (1981-2009), Sat=55% - Synthetic		
Simulation variant :		New simulation variant				
		Simulation date	19/06/26 15h14			
Simulation parameters						
Collector Plane Orientation		Tilt	25°	Azimuth	0°	
Models used		Transposition	Perez	Diffuse	Perez, Meteonorm	
Horizon		Free Horizon				
Near Shadings		No Shadings				
PV Arrays Characteristics (2 kinds of array defined)						
PV module		Si-mono	Model	STP555S-C72/Vmh		
Custom parameters definition		Manufacturer	Suntech			
Sub-array "Sub-array #1"						
Number of PV modules		In series	18 modules	In parallel	8 strings	
Total number of PV modules		Nb. modules	144	Unit Nom. Power	555 Wp	
Array global power		Nominal (STC)	79.9 kWp	At operating cond.	73.5 kWp (50°C)	
Array operating characteristics (50°C)		U mpp	691 V	I mpp	106 A	
Sub-array "Sub-array #2"						
Number of PV modules		In series	18 modules	In parallel	8 strings	
Total number of PV modules		Nb. modules	144	Unit Nom. Power	555 Wp	
Array global power		Nominal (STC)	79.9 kWp	At operating cond.	73.5 kWp (50°C)	
Array operating characteristics (50°C)		U mpp	691 V	I mpp	106 A	
Total Arrays global power		Nominal (STC)	160 kWp	Total	288 modules	
		Module area	744 m²			
Inverter						
Custom parameters definition		Model	Growatt MAX 75KTL3 LV			
		Manufacturer				
Characteristics		Operating Voltage	200-1000 V	Unit Nom. Power	75 kWac	
Sub-array "Sub-array #1"		Nb. of inverters	1 units	Total Power	75 kWac	
Sub-array "Sub-array #2"		Nb. of inverters	1 units	Total Power	75 kWac	
Total		Nb. of inverters	2	Total Power	150 kWac	
PV Array loss factors						
Thermal Loss factor		Uc (const)	20.0 W/m²K	Uv (wind)	0.0 W/m²K / m/s	
Wiring Ohmic Loss		Array#1	105 mOhm	Loss Fraction	1.5 % at STC	
		Array#2	105 mOhm	Loss Fraction	1.5 % at STC	
		Global		Loss Fraction	1.5 % at STC	
Module Quality Loss				Loss Fraction	3.0 %	
Module Mismatch Losses				Loss Fraction	1.0 % at MPP	
Incidence effect, ASHRAE parametrization		IAM =	1 - bo (1/cos i - 1)	bo Param.	0.05	

Grid-Connected System: Simulation parameters (continued)

User's needs :

Unlimited load (grid)

Grid-Connected System: Main results

Project : Graduation project

Simulation variant : New simulation variant

Main system parameters

PV Field Orientation

PV modules

PV Array

Inverter

Inverter pack

User's needs

System type

tilt

Model

Nb. of modules

Model

Nb. of units

Unlimited load (grid)

Grid-Connected

25°

azimuth

0°

Pnom 555 Wp

Pnom total **160 kWp**

Pnom 75.0 kW ac

Pnom total **150 kW ac**

Main simulation results

System Production

Produced Energy

296040 kWh/year

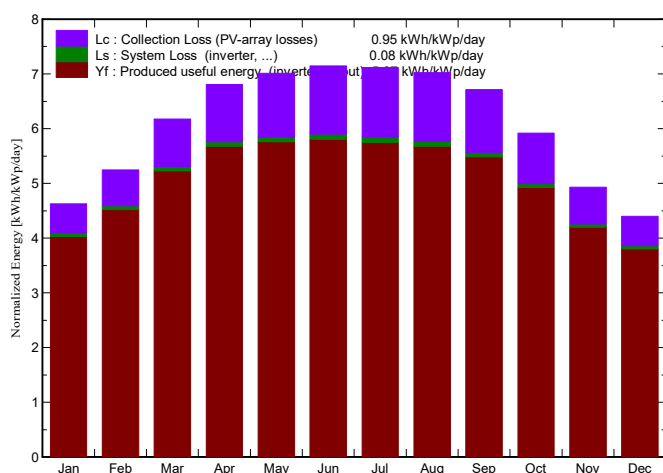
Specific prod.

1852 kWh/kWp/year

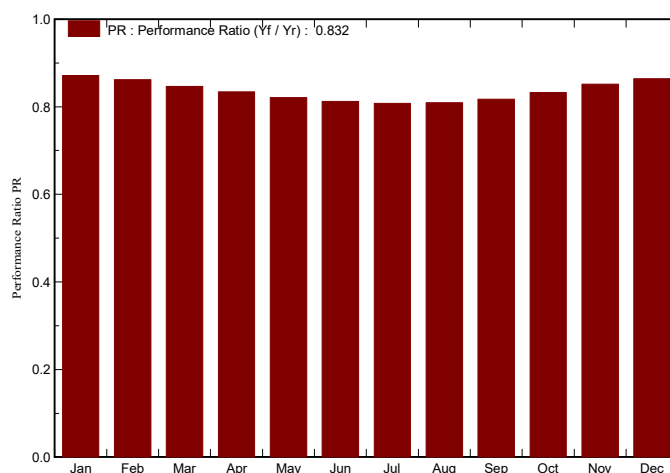
Performance Ratio PR

83.2 %

Normalized productions (per installed kWp): Nominal power 160 kWp



Performance Ratio PR



New simulation variant Balances and main results

	GlobHor	T Amb	GlobInc	GlobEff	EArray	E_Grid	EffArrR	EffSysR
	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	%	%
January	102.1	13.13	143.5	139.5	20277	19978	18.99	18.71
February	115.4	14.14	146.9	143.1	20539	20235	18.80	18.52
March	166.7	17.26	191.5	186.5	26304	25907	18.46	18.18
April	195.3	20.20	204.2	198.3	27623	27218	18.18	17.92
May	225.4	24.03	217.4	210.9	28974	28538	17.91	17.64
June	232.6	26.55	214.4	207.9	28284	27839	17.73	17.45
July	235.0	28.26	220.7	214.0	28942	28491	17.63	17.35
August	216.8	28.06	217.7	211.4	28566	28152	17.64	17.38
September	181.1	25.96	201.4	196.0	26696	26304	17.82	17.56
October	148.7	23.04	183.5	178.8	24763	24405	18.14	17.88
November	109.7	18.39	147.9	144.0	20427	20131	18.57	18.30
December	95.2	14.83	136.4	132.9	19139	18843	18.85	18.56
Year	2023.9	21.19	2225.4	2163.3	300534	296040	18.15	17.88

Legends: GlobHor

Horizontal global irradiation

T Amb

Ambient Temperature

GlobInc

Global incident in coll. plane

GlobEff

Effective Global, corr. for IAM and shadings

EArray

Effective energy at the output of the array

E_Grid

Energy injected into grid

EffArrR

Effic. Eout array / rough area

EffSysR

Effic. Eout system / rough area

Grid-Connected System: Loss diagram

Project :Graduation project

Simulation variant :New simulation variant

Main system parameters	System type	Grid-Connected		
PV Field Orientation	tilt	25°	azimuth	0°
PV modules	Model	STP555S-C72/Vmh	Pnom	555 Wp
PV Array	Nb. of modules	288	Pnom total	160 kWp
Inverter	Model	Growatt MAX 75KTL3 LV	Pnom	75.0 kW ac
Inverter pack	Nb. of units	2.0	Pnom total	150 kW ac
User's needs	Unlimited load (grid)			

Loss diagram over the whole year

